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jax-smfsb: A python library for stochastic systems biology modelling and inference

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Software

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Summary

Many biological processes, and especially molecular biochemical processes, exhibit non-trivial stochasticity in their dynamical behaviour (Wilkinson, 2009). The popular textbook *Stochastic modelling for systems biology, third edition* (Wilkinson, 2018) describes the stochastic approach to modelling and simulation of biochemical processes, and how to do Bayesian inference for the parameters of such models using time course data (Golightly & Wilkinson, 2011). `jax-smfsb` provides a fast and efficient implementation of all of the algorithms described in Wilkinson (2018), able to effectively exploit multiple cores and GPUs, leading to performance suitable for the analysis of non-trivial research problems.

Statement of Need

Although there exist many tools for modelling biological network dynamics using deterministic approaches, typically based on ordinary differential equations (ODEs), there are relatively few flexible software libraries for modelling and simulation of stochastic biochemical networks, although *libRoadRunner* (Welsh et al., 2022) and *SBSCL* (Panchiwala et al., 2021) are notable examples. There are even fewer libraries for principled (fully Bayesian) inference for the parameters of such networks using data.

In addition to describing the mathematical framework for stochastic modelling, simulation, and inference, Wilkinson (2018) also describes a software implementation of all of the algorithms. The language chosen to illustrate the implementation was R (R Core Team, 2024), and the library is available as the package `smfsb` (Wilkinson, 2024). While this library is of significant pedagogical value, the overheads of dynamic interpreted languages such as R make it unsuitable for the development of high-performance codes needed for non-trivial research problems. An implementation in the compiled strongly-typed functional language Scala (Odersky et al., 2004), `scala-smfsb` (Wilkinson, 2019) partially addresses this issue, but the lack of systems biology students and researchers familiar with Scala has limited the impact of this library. More recently, a Python (Van Rossum & Drake, 2009) port of the library, `python-smfsb` (Wilkinson, 2023) has been developed, utilising the Python libraries `numpy` (Harris et al., 2020) and `scipy` (Virtanen et al., 2020). This is of significant pedagogical value, since Python has become a more popular programming language for systems biology modelling than R. Nevertheless, the performance of this library is similar to that of the R library, inadequate for serious research problems.

`jax-smfsb` addresses all of the limitations of the previously described implementations. It is essentially a port of `python-smfsb` with `numpy` and `scipy` replaced by JAX (Bradbury et al., 2018). JAX is a state-of-the-art high-performance machine learning framework that turns out to be well-suited to a range of problems in numerical, scientific and statistical computing. JAX is effectively a functional language for differentiable array processing embedded in Python, allowing just-in-time compilation and execution on modern hardware with state-of-the-art

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performance. In addition to a large number of machine learning libraries based on JAX, a growing ecosystem of libraries for scientific computing is developing; see, for example, `diffrax` (Kidger, 2021), JAX-MD (Schoenholz & Cubuk, 2021), and JAX-Fluids (Bezgin et al., 2023). `jax-smfsb` adds to this ecosystem by providing tools for modelling, simulation and Bayesian inference for stochastic (biochemical) network models.

Features

In addition to providing a small library of pre-defined models, and allowing direct specification of models as stochastic Petri nets using JAX arrays, `jax-smfsb` can also parse models encoded in the Systems Biology Markup Language (SBML) (Keating et al., 2020) using `libSBML` (Bornstein et al., 2008), in addition to those encoded using the SBML-shorthand notation used in Wilkinson (2018). Exact and approximate stochastic simulation algorithms are provided for both the well-mixed and spatial (reaction-diffusion) case. Exact and approximate Bayesian inference algorithms for parameter inference are provided based on ABC (Marin et al., 2011), ABC-SMC (Toni et al., 2008) or particle MCMC (Andrieu et al., 2010) approaches.

Many demos are provided in the `demos` directory. These are direct equivalents to the demos provided in the `python-smfsb` library, so it is easy to compare the performance of the Python+JAX implementations to the more conventional Python+Numpy implementation. Precise timings are very hardware dependent, but on a high-specification machine, speed-ups of around two orders of magnitude for computationally intensive simulation or inference tasks can be expected.

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